CLASS XI & XII · CODE 840

APPLIED MATHS

FORMULA SHEET

Formulas · Charts · Graphs · Important Points

by

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Compiled from the CBSE Applied Mathematics (Code 840) Students' Handbook, Class XI & XII.

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Inside: every important formula, worked-out charts & graphs (histogram, pie chart, ogive, Venn diagrams, function graphs, normal curve, LP region and more), key points and exam tips — Class XI and Class XII.

CLASS XI

1

Number Theory

Prime numbers · Ratio & proportion · Logarithms

8 marks

PRIME NUMBERS

- **Prime**: exactly two factors (1 and itself). **2** is the only even prime; **1** is neither prime nor composite.
- Primes up to 50: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.
- **Fundamental Theorem of Arithmetic**: every integer > 1 is a product of primes, uniquely (apart from order).
- **Euclid**: there are infinitely many primes. **Twin primes** differ by 2: (3,5), (11,13), (17,19).
- **Bertrand's postulate**: for every n > 1 there is a prime p with n < p < 2n.
- **Sieve of Eratosthenes**: to list primes up to N, strike out multiples of every prime ≤ √N. The same test checks if n is prime: no prime ≤ √n divides it.
- **Ramanujan primes**: R_1=2, R_2=11, R_3=17, R_4=29, ...
- **Encryption (RSA idea)**: pick two huge primes p, q; publish n = pq. Multiplying is easy, factorising n is extremely hard, so the message stays secure.

Prime counting	$\pi(n) \approx \dfrac{n}{\ln n}$
Fermat's little thm	$a^{p-1} \equiv 1 \pmod{p}$ $(p \text{ prime},\ \gcd(a,p)=1)$
Mersenne number	$M_p = 2^p - 1$ $(p \text{ prime})$

RATIO & PROPORTION

Ratio	$a:b = \dfrac{a}{b}$ $(b \neq 0)$
Proportion	$a:b::c:d \Rightarrow ad = bc$
Mean proportional	$b^2 = ac \Rightarrow b = \sqrt{ac}$
Third proportional	$a:b = b:c \Rightarrow c = \dfrac{b^2}{a}$
Componendo	$\dfrac{a}{b} = \dfrac{c}{d} \Rightarrow \dfrac{a+b}{b} = \dfrac{c+d}{d}$
Dividendo	$\dfrac{a-b}{b} = \dfrac{c-d}{d}$
Comp.-dividendo	$\dfrac{a+b}{a-b} = \dfrac{c+d}{c-d}$

Ratio	Name
a² : b²	Duplicate ratio
√a : √b	Sub-duplicate ratio
a³ : b³	Triplicate ratio
ac : bd	Compound of a:b and c:d

- **Partnership**: profit shared in the ratio (capital × time) of each partner.
- **Mixtures / sharing**: if a sum is divided in a : b, shares are a/(a+b) and b/(a+b) of the total.

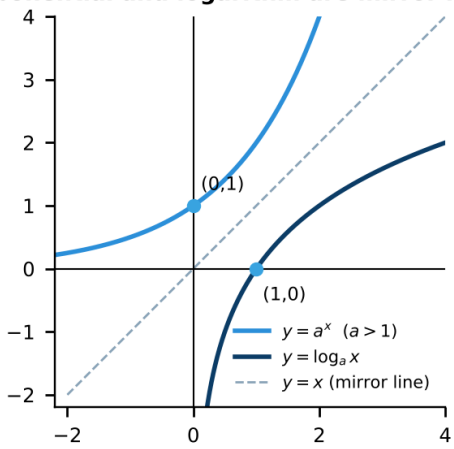
LOGARITHMS & ANTILOGARITHMS

Definition	$\log_a x = y \Leftrightarrow a^y = x$ $(a>0,\ a\neq 1,\ x>0)$
Product rule	$\log_a(mn) = \log_a m + \log_a n$
Quotient rule	$\log_a \dfrac{m}{n} = \log_a m - \log_a n$
Power rule	$\log_a m^n = n\log_a m$
Change of base	$\log_a b = \dfrac{\log_c b}{\log_c a} = \dfrac{1}{\log_b a}$
Special values	$\log_a 1 = 0,\ \log_a a = 1,\ a^{\log_a x} = x$
Common & natural	$\log = \log_{10},\ \ln = \log_e,\ e \approx 2.718$
Antilog	$\log x = y \Rightarrow x = \text{antilog}(y) = 10^y$

- **Characteristic** = integer part of log N; **mantissa** = decimal part (always 0 ≤ m < 1).
- N > 1: characteristic = (digits before the decimal point) − 1. 0 < N < 1: characteristic = −(zeros after the point + 1), written with a bar.

TIP: log 2 = 0.3010, log 3 = 0.4771, log 5 = 0.6990, log 7 = 0.8451. Use log 10 = 1 and the rules to get others (log 6 = log 2 + log 3).

Exponential and logarithm are mirror images



Exponential $y = a^x$ and logarithm $y = \log_a x$ reflect in the line $y = x$

KEY POINTS TO REMEMBER

- log of a product = **sum**; log of a quotient = **difference**; log of a power = **multiplier**.
- log of a negative number or of 0 is not defined.
- A prime has exactly 2 factors; test divisibility only by primes up to √n.

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CLASS XI

2

Interpretation of Data

Frequency tables · Histogram · Pie chart · Ogive

8 marks

FREQUENCY DISTRIBUTION

Range	$\text{Range} = L - S$ (largest − smallest)
Class width	$h =$ upper limit − lower limit
Class mark	$x_i = \dfrac{\text{lower} + \text{upper}}{2}$
No. of classes	$\approx \dfrac{\text{Range}}{\text{class width}}$
Relative frequency	$\dfrac{f}{N}$, $\%\ \text{freq} = \dfrac{f}{N} \times 100$
Cumulative freq.	$cf_i = f_1 + f_2 + \cdots + f_i$, $cf_{last} = N = \sum f$

- **Inclusive** classes (10–19, 20–29) → make **exclusive** by subtracting 0.5 from every lower limit and adding 0.5 to every upper limit (true class limits).
- Tally marks are counted in bundles of five; check that Σ f = N.

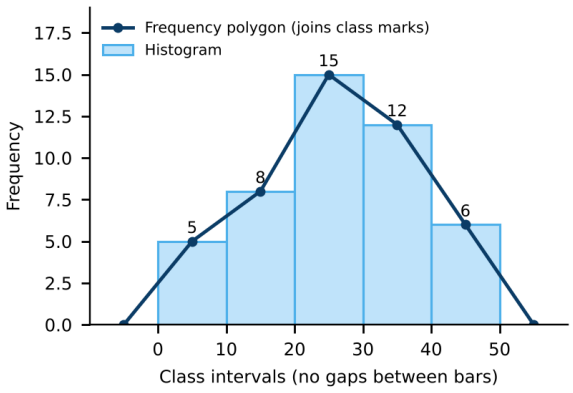
CHOOSING THE RIGHT GRAPH

Graph	Used for	How it is drawn
Bar chart	Categories	Equal gaps; height = value
Histogram	Continuous classes	No gaps; area ∝ f
Freq. polygon	Shape of data	Join class marks (add 0-f ends)
Ogive	Cumulative data	Plot cf vs **upper** limit (less-than) / lower limit (more-than)
Pie chart	Parts of a whole	Angle = value/total × 360°
Line graph	Trend over time	Join points in time order

Pie angle	$\theta = \dfrac{\text{value}}{\text{total}} \times 360^\circ$
Unequal widths	$\text{height} \propto \dfrac{f}{h}$ (histogram)

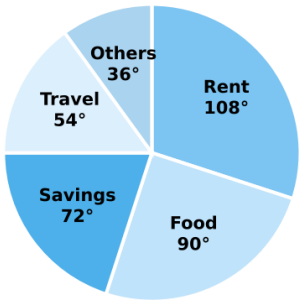
TIP: Median = x-coordinate of the point where the less-than and more-than ogives cross; its y-value is N/2.

Histogram & frequency polygon



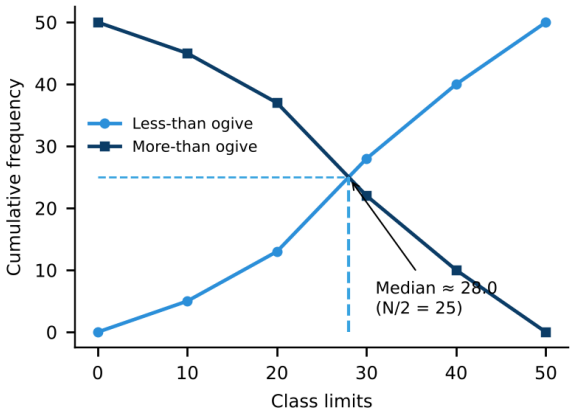
Histogram bars touch; the polygon joins the class marks and closes on the x-axis

Pie chart: angle = (value / total) × 360°



Pie chart: all angles add to 360°

Ogives: crossing point gives the median



Less-than and more-than ogives meet at the median

KEY POINTS TO REMEMBER

- Bar chart has gaps; histogram does **not**.
- Class mark is the middle value of a class.
- Pie angles must total 360°; percentages total 100%.
- Read graphs carefully: check axis scale, units and title before answering.

CLASS XI

3

Analysis of Data

Mean · Median · Mode · Dispersion · Skewness

10 marks

ARITHMETIC MEAN

Ungrouped	$\bar{x} = \dfrac{\sum x}{n}$
Direct (grouped)	$\bar{x} = \dfrac{\sum fx}{\sum f}$
Assumed mean	$\bar{x} = A + \dfrac{\sum fd}{\sum f}, \quad d = x - A$
Step deviation	$\bar{x} = A + h\dfrac{\sum fu}{\sum f}, \quad u = \dfrac{x - A}{h}$
Combined mean	$\bar{x} = \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

- $\Sigma(x - \bar{x}) = 0$. Adding k to every value adds k to the mean; multiplying by k multiplies the mean by k.

MEDIAN & MODE

Median (odd n)	$\text{Md} = \left(\dfrac{n+1}{2}\right)^{\text{th}}$ value
Median (even n)	$\text{Md} = \dfrac{(n/2)^{th} + (n/2+1)^{th}}{2}$
Median (grouped)	$\text{Md} = l + \dfrac{\frac{N}{2} - cf}{f} \times h$
Mode (grouped)	$\text{Mo} = l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
Empirical relation	$\text{Mo} = 3\,\text{Md} - 2\,\bar{x}$
Quartiles	$Q_1 = \left(\dfrac{n+1}{4}\right)^{th}, \quad Q_3 = 3\left(\dfrac{n+1}{4}\right)^{th}$

- In median/mode formulas: l = lower limit of the median/modal class, cf = cumulative freq. before it, f = its frequency, f_1 = modal-class freq., f_0, f_2 = freq. before / after, h = class width.

TIP: Median class is the first class whose cf ≥ N/2. Always arrange data in order (and use exclusive classes) before applying formulas.

GEOMETRIC & HARMONIC MEAN

Geometric mean	$\text{GM} = (x_1 x_2 \cdots x_n)^{1/n}$
GM using logs	$\log \text{GM} = \dfrac{\sum \log x}{n}$; $\dfrac{\sum f \log x}{N}$
Harmonic mean	$\text{HM} = \dfrac{n}{\sum \frac{1}{x}}$; $\dfrac{N}{\sum \frac{f}{x}}$
Two numbers a, b	$\text{HM} = \dfrac{2ab}{a+b}, \quad \text{GM} = \sqrt{ab}$
Order of means	$\text{AM} \geq \text{GM} \geq \text{HM}$
Two numbers	$\text{GM}^2 = \text{AM} \times \text{HM}$

MEASURES OF DISPERSION

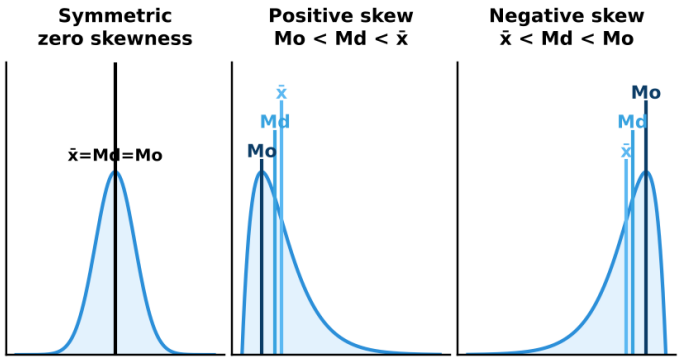
Range	$L - S, \quad \text{coeff.} = \dfrac{L-S}{L+S}$
Quartile deviation	$QD = \dfrac{Q_3 - Q_1}{2}, \quad \text{coeff.} = \dfrac{Q_3 - Q_1}{Q_3 + Q_1}$
Mean deviation	$\text{MD} = \dfrac{\sum \lvert x - a \rvert}{n}$; $\dfrac{\sum f \lvert x - a \rvert}{\sum f}$
Coeff. of MD	$\dfrac{\text{MD}}{a}$ (a = mean or median)
Variance	$\sigma^2 = \dfrac{\sum (x - \bar{x})^2}{n} = \dfrac{\sum x^2}{n} - \bar{x}^2$
Variance (grouped)	$\sigma^2 = \dfrac{\sum fx^2}{N} - \left(\dfrac{\sum fx}{N}\right)^2$
Standard deviation	$\sigma = \sqrt{\text{variance}}$
Step-deviation SD	$\sigma = h\sqrt{\dfrac{\sum fu^2}{N} - \left(\dfrac{\sum fu}{N}\right)^2}$
Coeff. of variation	$CV = \dfrac{\sigma}{\bar{x}} \times 100$

- Adding a constant to all values leaves SD unchanged; multiplying by k multiplies SD by |k|.
- **Lower CV = more consistent / stable** data; higher CV = more variable.

SKEWNESS

Karl Pearson	$Sk = \dfrac{\bar{x} - \text{Mo}}{\sigma} = \dfrac{3(\bar{x} - \text{Md})}{\sigma}$
Bowley	$Sk_B = \dfrac{Q_3 + Q_1 - 2\,\text{Md}}{Q_3 - Q_1}$

- Sk > 0 → positively skewed (long right tail); Sk < 0 → negatively skewed; Sk = 0 → symmetric.



Position of mean, median and mode in symmetric and skewed data

KEY POINTS TO REMEMBER

- **SD² = variance**; SD is in the same units as the data, CV has no units.
- Mean is affected by extreme values; median is not.
- Modal class has the highest frequency.
- Check: AM ≥ GM ≥ HM for positive numbers.

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CLASS XI

4

Commercial Mathematics

Profit & loss · Interest · Annuity · Depreciation

12 marks

PROFIT & LOSS

Profit / Loss	$\text{Profit} = SP - CP \;\; ; \;\; \text{Loss} = CP - SP$
Profit %	$\frac{\text{Profit}}{CP} \times 100$
Loss %	$\frac{\text{Loss}}{CP} \times 100$
SP from CP	$SP = CP\left(\frac{100 + P\%}{100}\right) \;;\; CP\left(\frac{100 - L\%}{100}\right)$
CP from SP	$CP = SP \times \frac{100}{100 + P\%} \;;\; SP \times \frac{100}{100 - L\%}$
Discount	$\text{Disc} = MP - SP, \quad \text{Disc}\% = \frac{\text{Disc}}{MP} \times 100$
Two discounts	$\text{net } \% = d_1 + d_2 - \frac{d_1 d_2}{100}$

- Profit % and loss % are always on the **cost price**; discount % is on the **marked price**. Overheads (transport, repairs) are added to CP.
- Two articles sold at the same price, one at x% gain and the other at x% loss → overall **loss of (x/10)%**. Equal CP with x% gain and x% loss → no profit, no loss.

SIMPLE & COMPOUND INTEREST

Simple interest	$SI = \frac{Prn}{100}, \quad A = P + SI = P\left(1 + \frac{rn}{100}\right)$
Compound (yearly)	$A = P\left(1 + \frac{r}{100}\right)^n, \quad CI = A - P$
Compounded m times	$A = P\left(1 + \frac{r}{100m}\right)^{mn}$
Effective rate	$E = \left(1 + \frac{r}{m}\right)^m - 1 \;\; (r \text{ in decimal})$
Rule of 72	$\text{Doubling time} \approx \frac{72}{r\%} \text{ years}$

- Half-yearly: rate r/2, periods 2n. Quarterly: rate r/4, periods 4n. Rate given for compounding more than once a year is the **nominal rate**.
- For 1 year: CI = SI. For 2 years: CI − SI = P(r/100)².

PRESENT VALUE, NPV & ANNUITY

Future value	$FV = PV(1 + r)^n$
Present value	$PV = \frac{FV}{(1 + r)^n} \quad \text{(discounting)}$
Net present value	$NPV = \sum_{t=1}^{n} \frac{C_t}{(1 + r)^t} - C_0$
Annuity (PV)	$PV = C\,\frac{1 - (1 + i)^{-n}}{i}$
Annuity (FV)	$FV = C\,\frac{(1 + i)^n - 1}{i}$
Perpetuity	$PV = \frac{C}{i}$

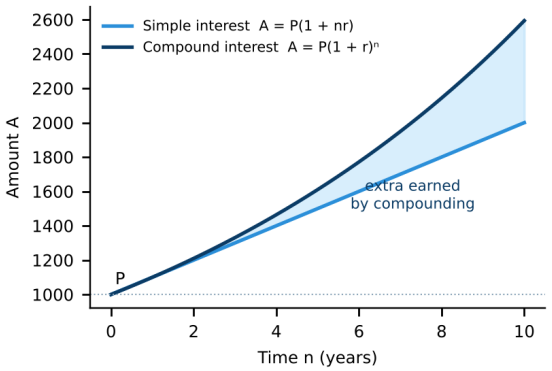
- Ordinary annuity: payments at the **end** of each period. Annuity due (payments at the start): multiply the above by (1 + i).
- **Accept** a project if NPV > 0; reject if NPV < 0. i = rate per period, n = number of periods.

DEPRECIATION

Straight line	$D = \frac{\text{Cost} - \text{Scrap value}}{\text{Life (years)}}$
Rate of dep.	$\frac{D}{\text{Original cost}} \times 100$
Reducing balance	$V_n = C\left(1 - \frac{r}{100}\right)^n$

- Original cost includes purchase price + carriage + installation/erection charges.
- Straight line: same depreciation every year. Reducing balance (WDV): depreciation falls each year.

Simple vs compound interest (P=1000, r=10%)



Compound interest grows exponentially; simple interest is a straight line

KEY POINTS TO REMEMBER

- P = principal, r = rate, n = time, A = amount.
- The rate and time must use the same period (monthly rate with months).
- Depreciation lowers value; scrap value is what is left at the end of life.

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CLASS XI

5

Set Theory

Sets · Venn diagrams · Set operations

8 marks

BASICS

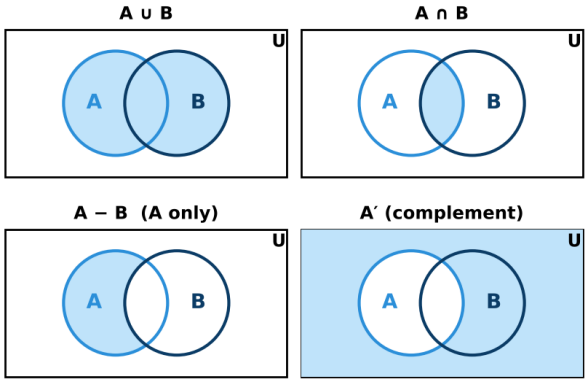
- **Roster form** {1, 2, 3}; **set-builder** {x : x is ...}. ∅ = empty set; U = universal set.
- **Finite / infinite** sets; **equal** sets have the same elements; **A ⊆ B** every element of A is in B; **A ⊂ B** proper subset.
- Intervals: (a, b) open, [a, b] closed, [a, b) half-open. ∅ is a subset of every set.
- **Union** A ∪ B (in A or B); **Intersection** A ∩ B (in both); **Difference** A − B (in A, not B); **Complement** A′ = U − A.
- **Disjoint** sets: A ∩ B = ∅. Symmetric difference A Δ B = (A − B) ∪ (B − A).

COUNTING FORMULAS

Two sets	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Three sets	$n(A \cup B \cup C) = n(A) + n(B) + n(C)$ $-n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$
Complement	$n(A') = n(U) - n(A)$
Only A	$n(A - B) = n(A) - n(A \cap B) = n(A \cap B')$
Neither A nor B	$n(A' \cap B') = n(U) - n(A \cup B)$
Exactly one of A,B	$n(A) + n(B) - 2\,n(A \cap B)$
Disjoint sets	$n(A \cup B) = n(A) + n(B)$
No. of subsets	2^n, proper subsets $= 2^n - 1$
Power set	$n(P(A)) = 2^{n(A)}$

LAWS OF SETS

De Morgan	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Commutative	$A \cup B = B \cup A$, $A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$
Idempotent	$A \cup A = A$, $A \cap A = A$
Identity	$A \cup \emptyset = A$, $A \cap U = A$
Complement laws	$A \cup A' = U$, $A \cap A' = \emptyset$, $(A')' = A$



Shaded regions of the four basic Venn diagrams

KEY POINTS TO REMEMBER

- Start Venn problems from the **innermost** region (A ∩ B ∩ C) and work outwards.
- A set with n elements has 2^n subsets.
- A − B and B − A are different sets; A − B = A ∩ B′.

CLASS XI

6

Relations & Functions

Cartesian product · Types of functions · Domain & range

5 marks

RELATIONS

Cartesian product	$A \times B = \{(a, b) : a \in A,\ b \in B\}$
Count of pairs	$n(A \times B) = n(A) \cdot n(B)$
Relations A → B	$2^{n(A)\,n(B)}$ (subsets of $A \times B$)

- A **relation** R from A to B is any subset of A × B. Domain = first elements; range = second elements; co-domain = B.
- **Reflexive**: (a,a) ∈ R for all a. **Symmetric**: (a,b) ∈ R ⇒ (b,a) ∈ R. **Transitive**: (a,b),(b,c) ∈ R ⇒ (a,c) ∈ R. **Equivalence** = all three.

FUNCTIONS

- A **function** f : A → B assigns **every** element of A **exactly one** image in B. Every function is a relation; not every relation is a function.
- **Vertical-line test**: a graph is a function if no vertical line cuts it twice. **Horizontal-line test**: one-one if no horizontal line cuts it twice.

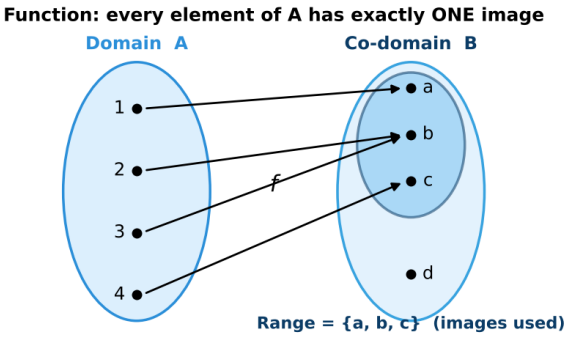
Type	Meaning
One-one	Different elements → different images
Many-one	Two or more elements share an image
Onto	Range = co-domain
Into	Range ⊂ co-domain (not equal)
Bijective	Both one-one and onto

Functions A→B	$[n(B)]^{n(A)}$
One-one functions	${}^{n(B)}P_{n(A)}$ $(n(A) \le n(B))$
Bijections	$n!$ $(n(A) = n(B) = n)$
Composite	$(f \circ g)(x) = f(g(x))$
Inverse	$f^{-1}(f(x)) = x$ (f must be bijective)

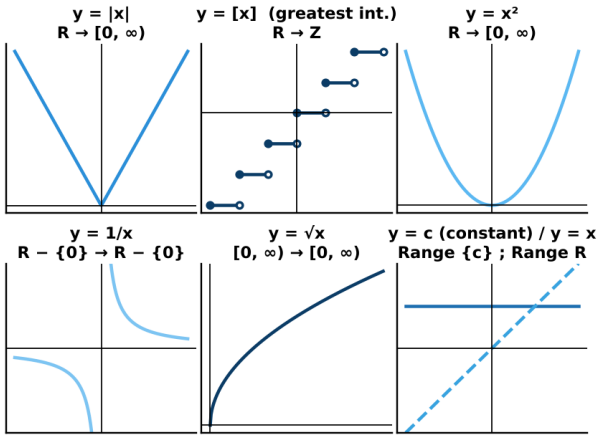
- In general f∘g ≠ g∘f. Domain of f ± g and fg is the common domain; for f/g exclude points where g = 0.

DOMAIN & RANGE OF STANDARD FUNCTIONS

f(x)	Domain	Range
c (constant)	R	{c}
x	R	R
|x|	R	[0, ∞)
x^2	R	[0, ∞)
x^3	R	R
1/x	R − {0}	R − {0}
√x	[0, ∞)	[0, ∞)
[x] (greatest int.)	R	Z (integers)



Domain, co-domain and range in an arrow diagram



Graphs of standard functions

KEY POINTS TO REMEMBER

- Range ⊆ co-domain; range = set of images actually used.
- Check the function definition: no element of A may be left without an image, and none may have two images.
- Denominator ≠ 0 and expression under √ ≥ 0 decide the domain.

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CLASS XI

7

Algebra

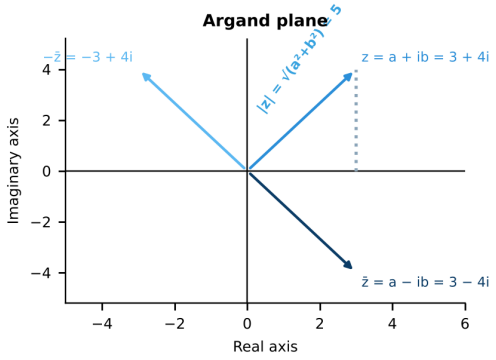
Complex numbers · Sequence & series · P&C · Binomial

14 marks

(A) COMPLEX NUMBERS

Imaginary unit	$i=\sqrt{-1},\quad i^2=-1,\quad \sqrt{-a}=i\sqrt{a}$
Powers of i	$i^{4k}=1,\ i^{4k+1}=i,\ i^{4k+2}=-1,\ i^{4k+3}=-i$
Form & modulus	$z=a+ib,\quad \lvert z\rvert=\sqrt{a^2+b^2}$
Conjugate	$\bar{z}=a-ib,\quad z\bar{z}=\lvert z\rvert^2$
Add / subtract	$(a+ib)\pm(c+id)=(a\pm c)+i(b\pm d)$
Multiply	$(a+ib)(c+id)=(ac-bd)+i(ad+bc)$
Divide	$\dfrac{a+ib}{c+id}=\dfrac{(a+ib)(c-id)}{c^2+d^2}$
Reciprocal	$\dfrac{1}{z}=\dfrac{\bar{z}}{\lvert z\rvert^2}=\dfrac{a-ib}{a^2+b^2}$
Quadratic, D < 0	$x=\dfrac{-b\pm i\sqrt{4ac-b^2}}{2a}$
Polar form	$z=r(\cos\theta+i\sin\theta),\quad \theta=\tan^{-1}\dfrac{b}{a}$

- a + ib = c + id ⇔ a = c and b = d. Complex roots of a real quadratic come in conjugate pairs.
- $|z_1z_2|=|z_1||z_2|$; $|z_1/z_2|=|z_1|/|z_2|$.



Argand plane: z, its conjugate and modulus

(B) SEQUENCES & SERIES

AP: n-th term	$a_n=a+(n-1)d$
AP: sum	$S_n=\dfrac{n}{2}[2a+(n-1)d]=\dfrac{n}{2}(a+l)$
Common difference	$d=a_n-a_{n-1}$
Arithmetic mean	$A=\dfrac{a+b}{2}$; n AMs between a,b : $d=\dfrac{b-a}{n+1}$
GP: n-th term	$a_n=ar^{n-1},\quad r=\dfrac{a_n}{a_{n-1}}$
GP: sum (r > 1)	$S_n=\dfrac{a(r^n-1)}{r-1}$
GP: sum (r < 1)	$S_n=\dfrac{a(1-r^n)}{1-r}$
Infinite GP	$S_\infty=\dfrac{a}{1-r},\quad \lvert r\rvert<1$
Geometric mean	$G=\sqrt{ab}$; n GMs : $r=\left(\dfrac{b}{a}\right)^{\frac{1}{n+1}}$
AM-GM	$A\geq G,\quad G^2=A\cdot H$
Σ n	$\sum n=\dfrac{n(n+1)}{2}$
Σ n²	$\sum n^2=\dfrac{n(n+1)(2n+1)}{6}$
Σ n³	$\sum n^3=\left[\dfrac{n(n+1)}{2}\right]^2$

- Three terms in AP: a − d, a, a + d. In GP: a/r, a, ar.
- If a, b, c are in AP then 2b = a + c; in GP then b^2 = ac.

(C) PERMUTATIONS & COMBINATIONS

Factorial	$n!=n(n-1)\cdots2\cdot1,\quad 0!=1$
Permutation	${}^nP_r=\dfrac{n!}{(n-r)!}$
Combination	${}^nC_r=\dfrac{n!}{r!\,(n-r)!}$
Relations	${}^nP_r=r!\ {}^nC_r,\quad {}^nC_r={}^nC_{n-r}$
Pascal's rule	${}^nC_r+{}^nC_{r-1}={}^{n+1}C_r$
Repetition	$\dfrac{n!}{p!\,q!\,r!}$ (like objects)
Repetition allowed	n^r arrangements of r from n
Circular	$(n-1)!$ (distinct objects)
Handshakes / lines	${}^nC_2$; diagonals $={}^nC_2-n$

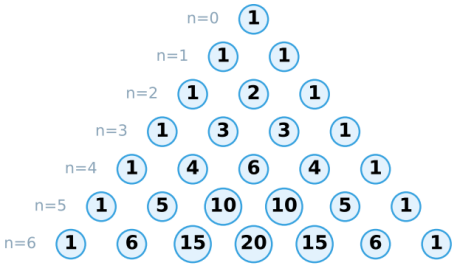
- **Multiplication principle**: m ways AND n ways → m × n. **Addition principle**: m ways OR n ways → m + n.
- Permutation = order matters (arrange); Combination = order does not matter (select).

BINOMIAL THEOREM

Expansion	$(a+b)^n=\sum_{r=0}^{n}{}^nC_r\,a^{n-r}b^r$
General term	$T_{r+1}={}^nC_r\,a^{n-r}b^r$
Middle term	n even : $T_{\frac{n}{2}+1}$; n odd : $T_{\frac{n+1}{2}},\ T_{\frac{n+3}{2}}$
Sum of coeffs.	$\sum{}^nC_r=2^n$
Even / odd sums	${}^nC_0+{}^nC_2+\cdots={}^nC_1+{}^nC_3+\cdots=2^{n-1}$

- $(a+b)^n$ has **n + 1** terms; powers of a fall from n to 0 while powers of b rise from 0 to n.

Pascal's triangle: binomial coefficients nCr



Each entry is the sum of the two above it

KEY POINTS TO REMEMBER

- $i^2=-1$; reduce high powers of i using the cycle of 4.
- |z| = distance of z from the origin; z·z̄ is always real and non-negative.
- AP: constant **difference**; GP: constant **ratio**.
- nCr counts selections; nPr counts arrangements.

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CLASS XI

8

Trigonometry

Identities · Compound angles · Heights & distances

5 marks

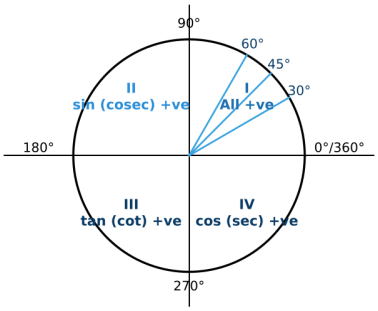
RATIOS & IDENTITIES

Radian measure	$\pi \text{ rad} = 180^\circ, \quad 1^\circ = \frac{\pi}{180} \text{ rad}$
Arc length	$l = r\theta$ (θ in radians)
Ratios	$\sin\theta = \frac{P}{H}, \ \cos\theta = \frac{B}{H}, \ \tan\theta = \frac{P}{B}$
Reciprocals	$\csc\theta = \frac{1}{\sin\theta}, \ \sec\theta = \frac{1}{\cos\theta}, \ \cot\theta = \frac{1}{\tan\theta}$
Quotient	$\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$
Pythagorean	$\sin^2\theta + \cos^2\theta = 1$
	$1 + \tan^2\theta = \sec^2\theta$
	$1 + \cot^2\theta = \csc^2\theta$

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	∞

- sin 15° = (√6 − √2)/4, cos 15° = (√6 + √2)/4, tan 15° = 2 − √3, tan 75° = 2 + √3.

Unit circle & signs of ratios (All–Sin–Tan–Cos)



ALLIED ANGLES (REDUCTION)

Angle	sin	cos	tan
−θ	−sin θ	cos θ	−tan θ
90° − θ	cos θ	sin θ	cot θ
90° + θ	cos θ	−sin θ	−cot θ
180° − θ	sin θ	−cos θ	−tan θ
180° + θ	−sin θ	−cos θ	tan θ
360° − θ	−sin θ	cos θ	−tan θ

TIP: Quadrant sign: **A**ll, **S**in, **T**an, **C**os (I, II, III, IV). Odd multiples of 90° swap sin↔cos, tan↔cot.

COMPOUND, DOUBLE & TRIPLE ANGLES

sin(A ± B)	$\sin A\cos B \pm \cos A\sin B$
cos(A ± B)	$\cos A\cos B \mp \sin A\sin B$
tan(A ± B)	$\frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$
sin 2A	$2\sin A\cos A = \frac{2\tan A}{1+\tan^2 A}$
cos 2A	$\cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
tan 2A	$\frac{2\tan A}{1-\tan^2 A}$
Half-angle forms	$\sin^2 A = \frac{1-\cos 2A}{2}, \quad \cos^2 A = \frac{1+\cos 2A}{2}$
Triple angle	$\sin 3A = 3\sin A - 4\sin^3 A, \quad \cos 3A = 4\cos^3 A - 3\cos A$

PRODUCT ↔ SUM

2 sin A cos B	$\sin(A+B) + \sin(A-B)$
2 cos A sin B	$\sin(A+B) - \sin(A-B)$
2 cos A cos B	$\cos(A+B) + \cos(A-B)$
2 sin A sin B	$\cos(A-B) - \cos(A+B)$
sin C + sin D	$2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$
sin C − sin D	$2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$
cos C + cos D	$2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$
cos C − cos D	$-2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$

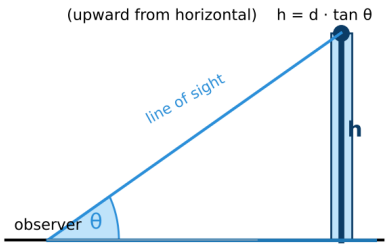
HEIGHTS & DISTANCES

Basic relation	$\tan\theta = \frac{h}{d} \Rightarrow h = d\tan\theta$
Two observations	$h = \frac{d}{\cot\beta - \cot\alpha} \quad (\alpha > \beta)$

- **Angle of elevation**: line of sight above the horizontal. **Angle of depression**: line of sight below the horizontal; it equals the angle of elevation from the other point.
- Draw a neat figure, mark known lengths/angles, pick the right-angled triangle, then use sin, cos or tan.

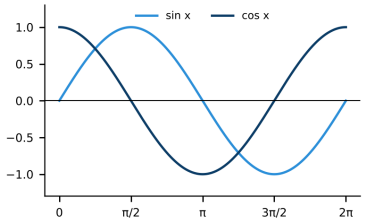
Heights & distances

Angle of elevation θ: tan θ = h / d



Angle of elevation of the top of a tower

Graphs of sin x and cos x (range −1 to 1, period 2π)



sin x and cos x repeat every 2π

KEY POINTS TO REMEMBER

- Learn the 0°, 30°, 45°, 60°, 90° table by heart.
- cos 2A has three forms — choose the one that matches the given data.
- Convert degrees ↔ radians with π = 180°.
- tan 90° and sec 90° are undefined.

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CLASS XII

1

Fundamentals of Calculus

Limits · Differentiation · Applications · Integration

14 marks

LIMITS & CONTINUITY

Definition	$\lim_{x\to a} f(x) = L \Leftrightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$
Limit laws	$\lim(f \pm g) = \lim f \pm \lim g, \quad \lim(fg) = \lim f \cdot \lim g$
Power limit	$\lim_{x\to a} \dfrac{x^n - a^n}{x - a} = n\,a^{n-1}$
Exponential	$\lim_{x\to 0} \dfrac{e^x - 1}{x} = 1, \quad \lim_{x\to 0} \dfrac{a^x - 1}{x} = \ln a$
Logarithmic	$\lim_{x\to 0} \dfrac{\ln(1+x)}{x} = 1$
Number e	$\lim_{x\to\infty} \left(1 + \dfrac{1}{x}\right)^x = \lim_{x\to 0} (1+x)^{1/x} = e$
At infinity	$\lim_{x\to\infty} \dfrac{1}{x^n} = 0 \quad (n > 0)$
Continuity at a	$\lim_{x\to a} f(x) = f(a)$

- If direct substitution gives 0/0: factorise and cancel, or rationalise. For ∞/∞ divide by the highest power of x.

DIFFERENTIATION

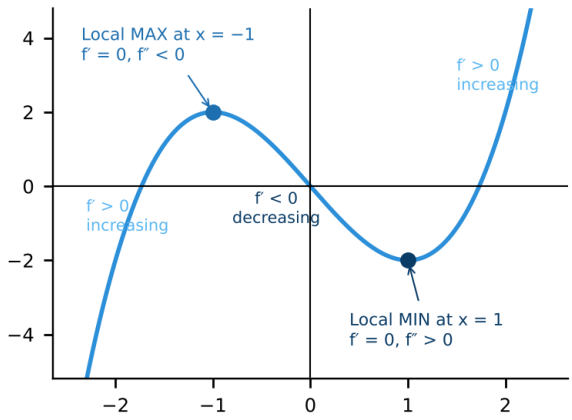
Definition	$f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$
Power rule	$\dfrac{d}{dx}(x^n) = n\,x^{n-1}, \quad \dfrac{d}{dx}(c) = 0$
Exponential	$\dfrac{d}{dx}e^x = e^x, \quad \dfrac{d}{dx}a^x = a^x \ln a$
Logarithm	$\dfrac{d}{dx}\ln x = \dfrac{1}{x}, \quad \dfrac{d}{dx}\log_a x = \dfrac{1}{x \ln a}$
Useful	$\dfrac{d}{dx}\sqrt{x} = \dfrac{1}{2\sqrt{x}}, \quad \dfrac{d}{dx}\dfrac{1}{x} = -\dfrac{1}{x^2}$
Sum / constant	$(cf \pm g)' = cf' \pm g'$
Product rule	$(uv)' = u'v + uv'$
Quotient rule	$\left(\dfrac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$
Chain rule	$\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$

APPLICATIONS OF DERIVATIVES

Marginal cost	$MC = \dfrac{dC}{dx}$
Marginal revenue	$MR = \dfrac{dR}{dx}$
Average cost	$AC = \dfrac{C(x)}{x}$
Revenue & profit	$R = p \cdot x, \quad \Pi = R - C$
Max profit	$MR = MC$ and $\dfrac{d^2\Pi}{dx^2} < 0$
Increasing	$f'(x) > 0$ (decreasing if $f'(x) < 0$)
Stationary points	$f'(c) = 0$
Second-derivative test	$f''(c) < 0 \Rightarrow \max, \quad f''(c) > 0 \Rightarrow \min$

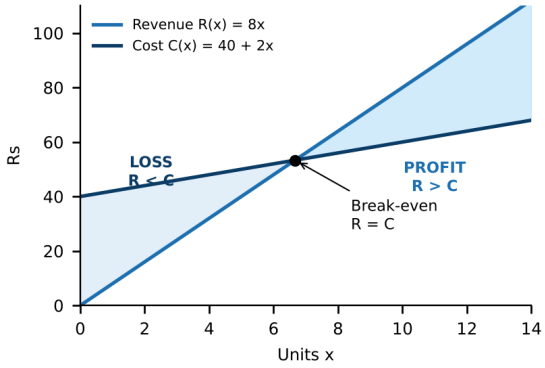
- **Steps for max/min**: (1) find f′(x); (2) solve f′(x) = 0; (3) find f″(x) at each root; (4) classify; (5) compute the max/min value from f(x).
- Marginal cost ≈ extra cost of producing **one more** unit.

f(x) = x³ − 3x : monotonicity & extrema



Sign of f′ decides rising/falling; f′ = 0 at turning points

Cost-revenue & break-even (profit = R − C)



Break-even where revenue equals cost

INTEGRATION

Power rule	$\int x^n dx = \dfrac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
Reciprocal	$\int \dfrac{1}{x} dx = \ln\lvert x\rvert + C$
Exponentials	$\int e^x dx = e^x + C, \quad \int a^x dx = \dfrac{a^x}{\ln a} + C$
Constant	$\int k\,dx = kx + C$
Linear argument	$\int (ax+b)^n dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$
Sum rule	$\int [f(x) \pm g(x)]dx = \int f dx \pm \int g\,dx$
Definite integral	$\int_a^b f(x)dx = F(b) - F(a)$
Cost from MC	$C(x) = \int MC\,dx + k \quad (k = \text{fixed cost})$
Revenue from MR	$R(x) = \int MR\,dx \quad (R(0) = 0)$

- Integration is the reverse of differentiation; always add the constant C for indefinite integrals.

KEY POINTS TO REMEMBER

- Marginal = derivative of the total function.
- Profit is maximum where **MR = MC** and the second derivative is negative.
- Fixed cost is the constant of integration for C(x).

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CLASS XII

2

Algebra

Matrices · Determinants (applications)

7 marks

MATRICES

- **Order** m × n means m rows and n columns.

Type	Property
Square	m = n
Diagonal	Non-diagonal entries are 0
Scalar	Diagonal with equal entries
Identity (I)	Diagonal entries all 1; AI = IA = A
Zero	All entries 0
Symmetric	$A^T = A$
Skew-symmetric	$A^T = -A$ (diagonal is 0)

Addition	$A \pm B$: same order, add/subtract entries
Scalar multiple	kA : multiply every entry by k
Product	$(AB)_{ij} = \sum_k a_{ik} b_{kj}$
Order of AB	$(m \times n)(n \times p) = m \times p$
Transpose	$(AB)^T = B^T A^T, \quad (A+B)^T = A^T + B^T$

- AB ≠ BA in general. AB exists only if (columns of A) = (rows of B). (AB)C = A(BC); A(B + C) = AB + AC.

2 × 2 determinant & inverse

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad |A| = ad - bc \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

3 × 3 determinant (expand along row 1)

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{matrix} a(ei-fh) \\ -\ b(di-fg) \\ +\ c(dh-eg) \end{matrix}$$

Sign pattern of cofactors: $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$A^{-1} = \frac{\text{adj}A}{|A|}$, $C_{ij} = (-1)^{i+j} M_{ij}$, adj A = (cofactor matrix)^T

Singular if |A| = 0 (no inverse)

Determinant and inverse formulas

DETERMINANT PROPERTIES

Product	$\lvert AB\rvert = \lvert A\rvert\,\lvert B\rvert, \quad \lvert A^T\rvert = \lvert A\rvert$
Scalar	$\lvert kA\rvert = k^n \lvert A\rvert$ (A is $n \times n$)
Inverse	$AA^{-1} = I, \quad (AB)^{-1} = B^{-1}A^{-1}$
Adjoint	$A\,(\text{adj}A) = \lvert A\rvert\, I, \quad \lvert\text{adj}A\rvert = \lvert A\rvert^{n-1}$

- Swapping two rows changes the sign of |A|. Two identical rows (or a zero row) → |A| = 0. Adding a multiple of one row to another does not change |A|.

APPLICATIONS OF DETERMINANTS

Area of triangle	$\Delta = \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$
Collinear points	$\Delta = 0$
Cramer's rule	$a_1x + b_1y = c_1, \quad a_2x + b_2y = c_2$
	$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}$
	$D = a_1b_2 - a_2b_1,\ D_x = c_1b_2 - c_2b_1,\ D_y = a_1c_2 - a_2c_1$

Matrix method	$AX = B \Rightarrow X = A^{-1}B \quad (\lvert A\rvert \neq 0)$

- D ≠ 0 → unique solution. D = 0 → no solution or infinitely many solutions (check D_x, D_y).

KEY POINTS TO REMEMBER

- Multiply matrices **row × column**.
- Inverse exists only when |A| ≠ 0.
- Area of a triangle is always taken as a positive value (use the modulus).

CLASS XII

3

Logical Reasoning

Series · Coding · Directions · Blood relations · Binary · Truth tables

8 marks

NUMBER SERIES & CODING-DECODING

Pattern	Look for
AP	Constant difference (+d)
GP	Constant ratio (×r)
Squares / cubes	n^2, n^3, $n^2 \pm 1$, $n^3 \pm n$
Primes	2, 3, 5, 7, 11, ...
Fibonacci	Each term = sum of previous two
Two series	Odd & even places follow separate rules
Difference of differences	Second differences constant

- **Letter positions**: A1 B2 C3 D4 E5 F6 G7 H8 I9 J10 K11 L12 M13 N14 O15 P16 Q17 R18 S19 T20 U21 V22 W23 X24 Y25 Z26.
- **Opposite letter**: position sum = 27 (A↔Z, B↔Y, M↔N). Check: forward shift (+k), backward shift (−k), reverse alphabet, position values, vowels/consonants substitution.

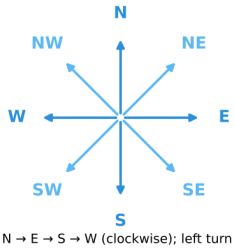
TIP: Decode with a known pair: compare each letter of the word with its code letter to find the rule, then apply to the new word.

DIRECTIONS

Shortest distance $d = \sqrt{x^2 + y^2}$ (right angle between paths)

- Right turn from N → E → S → W (clockwise); left turn is anticlockwise. A turn of 45° gives NE, SE, SW, NW.
- Morning: sun in the East, shadow falls West. Evening: shadow falls East. At noon the shadow falls towards the North (in the northern hemisphere).

Directions (each step = 45°; 90° between main directions)



Right turn from N → E → S → W (clockwise); left turn is anti-clockwise

Eight directions on a compass

BLOOD RELATIONS

Given relation	Called
Father's / mother's father	Grandfather
Father's brother	Paternal uncle
Mother's brother	Maternal uncle
Father's / mother's sister	Aunt
Brother's / sister's son	Nephew
Brother's / sister's daughter	Niece
Sister's husband; spouse's brother	Brother-in-law
Brother's wife; spouse's sister	Sister-in-law
Son's wife / daughter's husband	Daughter- / Son-in-law
Child of uncle or aunt	Cousin

TIP: Draw a family tree: □ / + for male, ○ / − for female, = for married couple, vertical line for parent–child.

SYLLOGISM

Statement	Type	Converts to
All A are B	Universal +ve	Some B are A
No A is B	Universal −ve	No B is A
Some A are B	Particular +ve	Some B are A
Some A are not B	Particular −ve	No conversion

Premises	Conclusion
All A are B + All B are C	All A are C
All A are B + No B is C	No A is C
Some A are B + All B are C	Some A are C
Some A are B + No B is C	Some A are not C
Two particular / two negative	No conclusion

- Use Venn diagrams: draw circles for the terms in the most general way and test each conclusion.

BINARY NUMBERS

Binary → decimal $(b_n b_{n-1} \ldots b_1 b_0)_2 = \sum b_k 2^k$

- **Decimal → binary**: divide repeatedly by 2; read the remainders from bottom to top.

Dec	0	1	2	3	4	5	6	7	8	9	10
Bin	0	1	10	11	100	101	110	111	1000	1001	1010

- **Addition**: 0+0 = 0, 0+1 = 1, 1+1 = 10 (write 0, carry 1), 1+1+1 = 11. Place values: ...32 16 8 4 2 1.
- 1's complement: flip every bit. 2's complement = 1's complement + 1 (used for subtraction).

LOGICAL OPERATIONS & TRUTH TABLES

p	q	¬p	p∧q	p∨q	p⊕q	p→q	p↔q
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q, \quad \neg(p \vee q) \equiv \neg p \wedge \neg q$
Implication	$p \to q \equiv \neg p \vee q$
Contrapositive	$p \to q \equiv \neg q \to \neg p$

- Converse of p → q is q → p; inverse is ¬p → ¬q. Only the **contrapositive** is always equivalent.
- **Tautology**: always true. **Contradiction**: always false. n statements → 2^n rows in a truth table.
- AND: true only if both true. OR: false only if both false. XOR: true if exactly one is true. NAND/NOR = negation of AND/OR.

KEY POINTS TO REMEMBER

- Positions of letters: A = 1 ... Z = 26; opposite letters add to 27.
- Draw a diagram for directions and blood relations before answering.
- Syllogism: two particular premises give no conclusion.
- Binary 1 + 1 = 10.

CLASS XII

4

Commercial Mathematics

EMI · Returns & CAGR · Stocks & bonds · GST · Banking

10 marks

EMI

Reducing balance	$EMI = \dfrac{P\,r(1+r)^n}{(1+r)^n - 1}$
Flat rate	$EMI = \dfrac{P(1+rt)}{12\,t}$ (r annual decimal, t years)
Total interest	$EMI \times n - P$

- In the reducing-balance formula r = annual rate ÷ 12 ÷ 100 (monthly, as a decimal) and n = number of months.
- Flat-rate interest is charged on the full original principal throughout, so it costs more than a reducing-balance loan at the same stated rate.

RETURNS & CAGR

Rate of return	$\dfrac{\text{Final} - \text{Initial} + \text{Income}}{\text{Initial}} \times 100$
CAGR	$CAGR = \left(\dfrac{V_n}{V_0}\right)^{\frac{1}{n}} - 1$
Real return	$\dfrac{1+\text{nominal}}{1+\text{inflation}} - 1 \approx \text{nominal} - \text{inflation}$

- V_0 = beginning value, V_n = ending value, n = number of years. CAGR smooths out year-to-year ups and downs.

STOCKS, SHARES & DEBENTURES

Dividend	$\dfrac{\text{rate}}{100} \times \text{Face value} \times \text{No. of shares}$
Dividend yield	$\dfrac{\text{Dividend per share}}{\text{Market value}} \times 100$
EPS	$\dfrac{\text{Net profit after tax}}{\text{No. of shares}}$
P/E ratio	$\dfrac{\text{Market price}}{EPS}$
No. of shares	$\dfrac{\text{Investment}}{\text{Market value (+ brokerage)}}$
Mutual fund NAV	$\dfrac{\text{Assets} - \text{Liabilities}}{\text{No. of units}}$

- Share at **par**: MV = FV; at **premium**: MV > FV; at **discount**: MV < FV. Dividend is always on **face value**.
- **Debenture**: a loan; fixed interest on face value (coupon rate), paid whether or not the company earns a profit.

BOND VALUATION

Value of a bond	$V = \sum_{t=1}^{n} \dfrac{C}{(1+r)^t} + \dfrac{F}{(1+r)^n}$
Current yield	$\dfrac{\text{Annual coupon}}{\text{Market price}} \times 100$

- C = coupon per period, F = face (maturity) value, r = required return per period. Bond price falls when market interest rates rise.

GST

GST amount	$GST = \text{Base price} \times \dfrac{g}{100}$
Invoice price	$\text{Base}\left(1 + \dfrac{g}{100}\right)$
Base from invoice	$\text{Base} = \text{Invoice} \times \dfrac{100}{100+g}$
Net GST payable	Output GST − Input tax credit

- **Within a state**: GST = CGST + SGST (each is half). **Between states**: IGST = full GST.

TIP: GST rate slabs get revised — always use the rate given in the question.

BANKING

Recurring deposit	$I = P \times \dfrac{n(n+1)}{2} \times \dfrac{r}{12 \times 100}$
Maturity value	$M = P\,n + I$ (n = months)
Fixed deposit (qtrly)	$A = P\left(1 + \dfrac{r}{400}\right)^{4t}$
Savings account	$I = \dfrac{\sum(\text{min balance 10th to month end}) \times r}{12 \times 100}$

- RD: P = monthly instalment. Savings interest uses the lowest balance between the 10th and the last day of each month.

KEY POINTS TO REMEMBER

- EMI uses a **monthly** rate and number of months.
- CAGR is not an arithmetic average of yearly returns.
- GST: input credit is subtracted from the GST collected.
- Dividend is on face value; yield is on market value.

CLASS XII

5

Probability

Rules · Total probability · Binomial · Poisson · Normal

10 marks

PROBABILITY RULES

Classical	$P(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}, \quad 0 \le P(E) \le 1$
Complement	$P(E') = 1 - P(E), \quad P(\text{at least one}) = 1 - P(\text{none})$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive	$P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B)$
Independent events	$P(A \cap B) = P(A)\,P(B)$
Conditional	$P(A\|B) = \dfrac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$
Multiplication rule	$P(A \cap B) = P(A)\,P(B\|A)$
Total probability	$P(A) = \sum_{i=1}^{n} P(E_i)\,P(A\|E_i)$
Bayes' theorem	$P(E_i\|A) = \dfrac{P(E_i)P(A\|E_i)}{\sum_j P(E_j)P(A\|E_j)}$

- $E_1,\ldots,E_n$ must be mutually exclusive and exhaustive (their union is the sample space).
- Sample-space facts: coin 2; die 6; two dice 36; pack of cards 52 (26 red, 26 black, 4 suits of 13, 12 face cards, 4 aces).

RANDOM VARIABLE

Mean (expectation)	$E(X) = \mu = \sum x_i p_i$
Variance	$\text{Var}(X) = \sum x_i^2 p_i - \mu^2, \quad \sigma = \sqrt{\text{Var}}$

- $\Sigma\, p_i = 1$ for every probability distribution.

BINOMIAL DISTRIBUTION

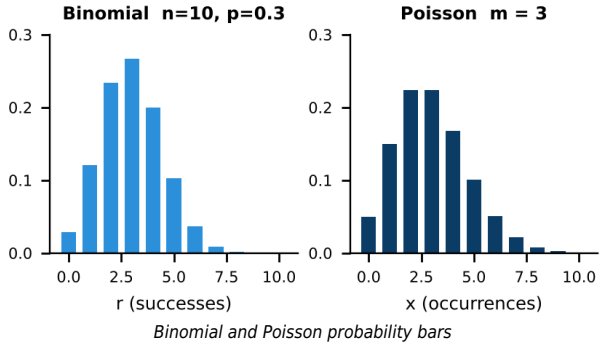
Probability	$P(X = r) = {}^{n}C_r\, p^r q^{n-r}, \quad q = 1 - p$
Mean	$\mu = np$
Variance & SD	$\sigma^2 = npq, \quad \sigma = \sqrt{npq}$

- Conditions: fixed number of trials n; each trial has only two outcomes (success/failure); trials are independent; p is constant.

POISSON DISTRIBUTION

Probability	$P(X = x) = \dfrac{e^{-m} m^x}{x!}, \quad x = 0, 1, 2, \ldots$
Mean = variance	$\mu = \sigma^2 = m$
From binomial	$m = np \;\; (n \text{ large}, \; p \text{ small})$

- Used for rare events in a fixed interval (defects per metre, calls per minute). $e^{-1} \approx 0.3679$, $e^{-2} \approx 0.1353$, $e^{-3} \approx 0.0498$.



Binomial and Poisson probability bars

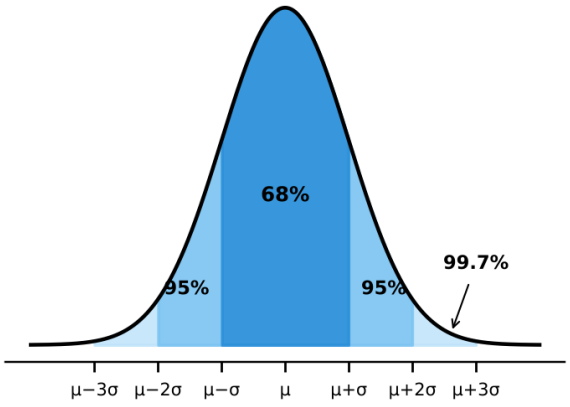
NORMAL DISTRIBUTION

Standard score	$z = \dfrac{x - \mu}{\sigma}$
Symmetry	$P(Z < -z) = 1 - P(Z < z)$

Interval	Area
μ ± 1σ	68.27 %
μ ± 2σ	95.45 %
μ ± 3σ	99.73 %
z = 0 (left half)	0.5000

- Bell-shaped, symmetric about μ; mean = median = mode; total area under the curve = 1; the tails never touch the axis.

Normal curve: mean = median = mode = μ



The empirical (68–95–99.7) rule

KEY POINTS TO REMEMBER

- P(A|B) divides by **P(B)**, the condition.
- Independent ≠ mutually exclusive.
- Poisson: mean = variance. Binomial: mean > variance.
- Convert to z before using the normal table.

By ThunderStudy

CLASS XII

6

Two-Dimensional Geometry

Slope · Equations of a line

4 marks

SLOPE OF A LINE

Slope	$m = \tan\theta = \dfrac{y_2 - y_1}{x_2 - x_1}$
Parallel lines	$m_1 = m_2$
Perpendicular	$m_1 m_2 = -1$
Angle between lines	$\tan\theta = \left\lvert \dfrac{m_2 - m_1}{1 + m_1 m_2} \right\rvert$
From Ax+By+C=0	$m = -\dfrac{A}{B}$

- Horizontal line: m = 0 (y = k). Vertical line: slope undefined (x = h). x-axis: y = 0; y-axis: x = 0.
- Three points are **collinear** if the slopes of AB and BC are equal.

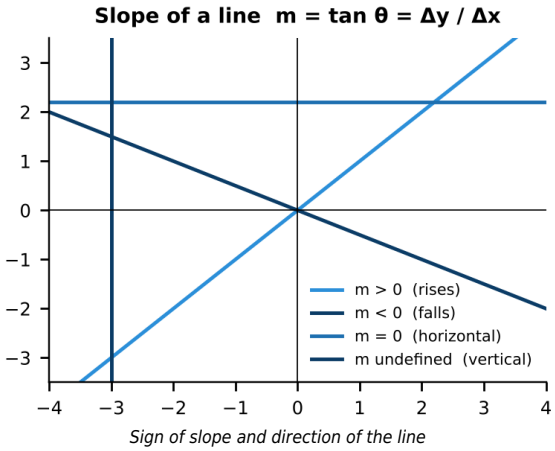
EQUATION OF A LINE

Point-slope	$y - y_1 = m(x - x_1)$
Slope–intercept	$y = mx + c$
Two-point	$y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
Intercept form	$\dfrac{x}{a} + \dfrac{y}{b} = 1$
General form	$Ax + By + C = 0$

- In y = mx + c, c is the y-intercept; the x-intercept is found by putting y = 0.

USEFUL COORDINATE FORMULAS

Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Mid-point	$\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$
Section (m : n)	$\left(\dfrac{mx_2 + nx_1}{m + n}, \dfrac{my_2 + ny_1}{m + n}\right)$
Point–line distance	$\dfrac{\lvert Ax_1 + By_1 + C\rvert}{\sqrt{A^2 + B^2}}$



Sign of slope and direction of the line

KEY POINTS TO REMEMBER

- Slope = rise ÷ run = tan θ (θ measured from the +x axis).
- Parallel → equal slopes; perpendicular → product −1.
- Choose the form of the equation that fits the data given.

CLASS XII

7

Linear Programming

Formulation · Graphical method · Corner-point method

10 marks

TERMINOLOGY

Term	Meaning
Decision variables	Quantities to be found (x, y)
Objective function	Linear function to maximise/minimise: Z = ax + by
Constraints	Linear inequalities from limited resources
Non-negativity	x ≥ 0, y ≥ 0
Feasible region	Common area satisfying all constraints
Feasible solution	Any point in the feasible region
Optimal solution	Feasible point giving the best Z

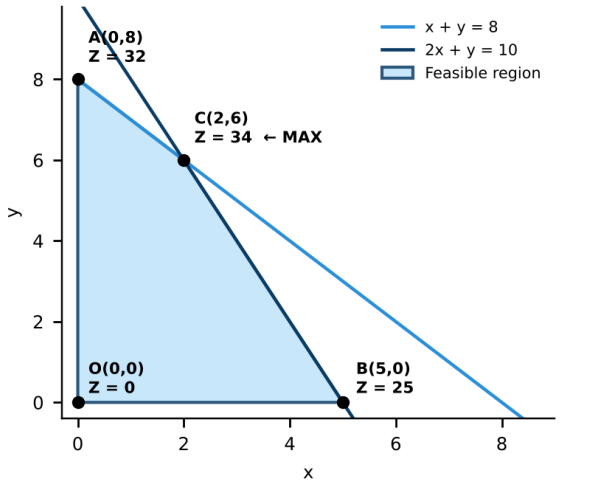
- Types: manufacturing (maximise profit), diet (minimise cost), transportation (minimise shipping cost).

GRAPHICAL METHOD — STEPS

- **1.** Define x, y; write the objective function Z and all constraints (with x, y ≥ 0).
- **2.** Draw each constraint as a line (replace ≤ / ≥ by =). Shade the correct side — test the point (0, 0).
- **3.** Mark the **feasible region** (common shaded area, first quadrant).
- **4.** Find the **corner points** (solve pairs of line equations at intersections).
- **5.** Evaluate Z at each corner point; the largest (max) or smallest (min) is the answer.
- **Corner-point theorem**: the optimum of Z occurs at a corner point of a bounded feasible region.
- If two corner points give the same optimum, every point on the line segment joining them is also optimal (multiple solutions).
- **Unbounded region**: a max/min may not exist; draw the half-plane Z > M (or < M) and check it has no common point with the region.
- No common region → problem is **infeasible**.

TIP: Always write the final answer with units, e.g. "Maximum profit = Rs 34 at x = 2, y = 6".

Max Z = 5x + 4y s.t. x+y ≤ 8, 2x+y ≤ 10, x,y ≥ 0



Feasible region with corner points; optimum at C(2, 6)

KEY POINTS TO REMEMBER

- Optimum is always at a **corner point**.
- Check every constraint when shading the region.
- Do not forget x ≥ 0 and y ≥ 0.

CLASS XII

8

Analysis of Time-Based Data

Index numbers · Consumer Price Index · Time series & trend

7 marks

INDEX NUMBERS

Price relative	$I = \frac{p_1}{p_0} \times 100$
Simple aggregate	$\frac{\sum p_1}{\sum p_0} \times 100$
Laspeyres (base wts)	$\frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$
Paasche (current wts)	$\frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$
Fisher (ideal)	$\sqrt{\text{Laspeyres} \times \text{Paasche}}$
Weighted avg. of relatives	$\frac{\sum IW}{\sum W}$

- p_0, q_0 = base-year price, quantity; p_1, q_1 = current-year price, quantity. Index of base year = 100.
- Uses: measuring inflation, adjusting wages (dearness allowance), comparing prices over time, policy making.

CONSUMER PRICE INDEX (CPI)

Aggregate expenditure	$CPI = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$
Family budget method	$CPI = \frac{\sum IW}{\sum W}$
Real wages	$\frac{\text{Money wages}}{CPI} \times 100$
Purchasing power	$\frac{100}{CPI} \times 100\ \%$
Inflation rate	$\frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \times 100$

- In the family budget method, W = weight = quantity or expenditure share of each item and I = price relative of the item.
- If CPI rises, purchasing power of money falls.

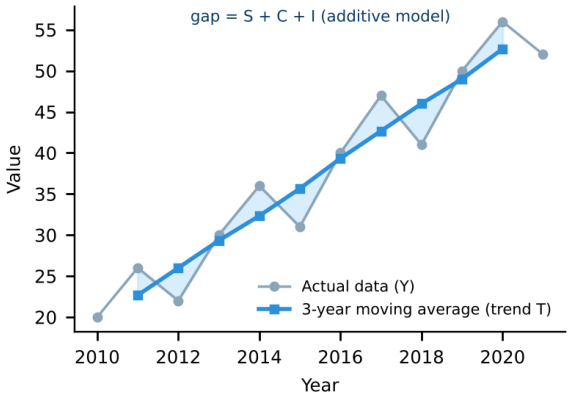
TIME SERIES & TREND

Component	Meaning
T – Trend	Long-term rise or fall
S – Seasonal	Regular change within a year (festivals, seasons)
C – Cyclical	Waves lasting more than a year (business cycle)
I – Irregular	Random, unpredictable (floods, strikes, war)

Additive model	$Y = T + S + C + I$
Multiplicative model	$Y = T \times S \times C \times I$
3-year moving avg.	$MA_t = \frac{Y_{t-1} + Y_t + Y_{t+1}}{3}$
5-year moving avg.	$MA_t = \frac{Y_{t-2} + \cdots + Y_{t+2}}{5}$

- Write each moving average against the **middle** year. With an odd period m you lose (m − 1)/2 values at each end.
- For an even period (e.g. 4), average each 4 values (it falls between two years), then average successive pairs to **centre** it.
- Trend values from the moving average; the remainder Y − T = S + C + I (additive model).

Time series: smoothing by moving average



The moving average smooths short-term ups and downs and shows the trend

KEY POINTS TO REMEMBER

- Base year index = 100.
- Laspeyres uses base-year quantities; Paasche uses current-year quantities.
- Moving averages smooth the series — they cannot be calculated for the end points.
- Higher CPI means higher cost of living.

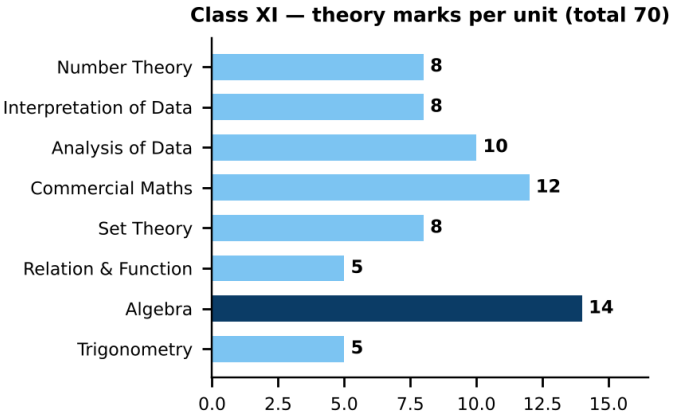
BONUS ★

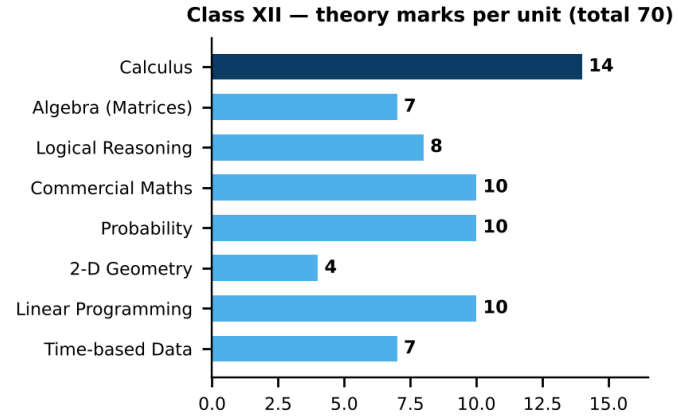
Exam Tips & Quick Reference

Important things to remember · marks weightage · constants

WHERE THE MARKS ARE

Both classes: **70 marks theory + 30 marks practical** (as per the handbook syllabus). Plan revision by weightage:





Theory marks per unit (handbook syllabus 2019–20)

TOP EXAM TIPS

- Write the **formula first**, then substitute — marks are given for the formula and steps.
- Keep **units** consistent (months vs years, % vs decimal) before using interest, EMI or annuity formulas.
- Draw a **neat labelled diagram**: Venn, histogram, ogive, LP region, heights & distances.
- For graphs: use a proper scale, label both axes, give the figure a title.
- Check answers: probability in [0, 1]; SD ≥ 0; |z| ≥ 0; sum of pie angles = 360°.
- In LP, test the corner points **and** state the final answer with values of x, y and Z.
- Use approximate calculator values only at the last step to avoid rounding errors.

COMMON MISTAKES TO AVOID

- Calculating profit/loss % on the **selling price** instead of the **cost price**.
- Forgetting the **+C** in indefinite integrals; forgetting to square-root the variance to get SD.
- Dividing by n instead of N or Σf in grouped data; using inclusive classes without correction.
- Mixing up nPr (order matters) and nCr (order doesn't).
- Confusing **mutually exclusive** events with **independent** events.
- Using nominal rate instead of the rate per period when compounding more than once a year.
- In determinants/matrices: multiplying element-by-element instead of row × column.

HANDY CONSTANTS & VALUES

Value	Number
π	3.1416 (22/7 ≈ 3.1429)
e	2.7183
√2, √3, √5	1.414, 1.732, 2.236
log 2, log 3	0.3010, 0.4771
log 5, log 7	0.6990, 0.8451
ln 10	2.3026
1 rad	57.3° (≈ 57°17′45″)

n	11	12	13	14	15	16	17	18	19	20
n^2	121	144	169	196	225	256	289	324	361	400

n	1	2	3	4	5	6	7	8	9	10
n^3	1	8	27	64	125	216	343	512	729	1000

LAST-MINUTE CHECKLIST

- Revise one unit per day: formulas → solved examples → previous-year questions.
- Focus first on the heavy units: Algebra (XI, 14), Commercial Maths (XI, 12), Calculus (XII, 14).
- Practice with timers; keep this sheet handy for last-minute revision.

SOURCE & NOTE

Compiled from the CBSE *Applied Mathematics (Subject Code 840) Students' Handbook* for Class XI and XII (session 2019–20 syllabus). Some additional standard results are included for completeness. Always cross-check with the latest syllabus from CBSE.